

Project Proposal

Project: AI-Powered Predictive Maintenance for Industrial Equipment

Objective:

The project aims to develop an AI-driven predictive maintenance system that continuously analyzes sensor data from industrial machines to forecast potential maintenance needs before failures occur. Additionally, the system will detect and report machine anomalies through acoustic signals in real time, enabling faster response and minimizing operational downtime.

Description:

This project integrates artificial intelligence and Internet of Things (IoT) technologies to collect, process, and analyze sensor data generated by industrial equipment. By monitoring machine health indicators such as vibration, temperature, pressure and acoustic signals, the system will assess equipment condition and identify anomalies and degradation patterns over time. This proactive approach helps prevent unexpected breakdowns, optimize maintenance scheduling, improve operational efficiency, and support data-driven decision-making in industrial environments.

Tools & Design: The project will be designed as a scalable AI-based predictive maintenance system that processes IoT sensor data through a structured pipeline including data preprocessing, feature engineering, machine learning modeling, and deployment. The architecture will support real-time prediction, model monitoring, and automated retraining using MLOps best practices.

Technologies to Use: Python, Pandas, NumPy, Scikit-learn, Librosa, TensorFlow/PyTorch, MLflow, FastAPI, and Azure Machine Learning.

Team Members & Responsibilities:

- 1- **Rowan Nasser** (Data Collection and Exploration - Data Visualization - Model Development and Optimization)
- 2- **Alaa Mohamed** (Data Collection, Exploration, and Preprocessing - Advanced Data Analysis and Feature Engineering - Model Development and Optimization)

- 3- **Nada Ahmed** (Data Collection, Exploration, and Preprocessing - Model Development and Optimization - MLOps)
 - 4- **Taghreed Mohammed** (Data Collection, Exploration, and Preprocessing - Advanced Data Analysis and Feature Engineering - Model Development and Optimization - MLOps)
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Week 1: *Data Collection, Exploration, and Preprocessing*

- **Data Collection (Responsible member(s): Alaa - Rowan - Nada - Taghreed)**
 - Collect sensor readings from industrial machinery, including signals such as vibration, temperature, and pressure. Moreover, collect machine acoustic recordings.
- **Data Exploration (Responsible member(s): Alaa - Rowan - Nada - Taghreed)**
 - Analyze the dataset to understand its structure, statistical distributions, and relationships among variables.
 - Detect and assess missing values, inconsistencies, noise, or abnormal data points that may affect modeling.
- **Data Preprocessing (Responsible member(s): Taghreed - Nada - Alaa)**
 - Treat missing data using appropriate techniques such as imputation or record removal.
 - Mitigate the impact of outliers and class imbalance through methods like resampling or synthetic data generation.
 - Engineer informative features, including rolling averages, frequency-domain attributes, and indicators of equipment degradation.

Deliverables:

- Dataset Exploration Report
 - Exploratory Data Analysis Notebook
 - Preprocessed Data
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Week 2: *Advanced Data Analysis and Feature Engineering*

- **Advanced Data Analysis (Responsible member(s): Taghreed - Alaa)**
 - Apply time-series analytical techniques to uncover trends, periodic behavior, and irregular patterns in sensor readings. Moreover, apply dimensionality reduction when necessary.
- **Feature Engineering (Responsible member(s): Taghreed - Alaa)**
 - Design additional predictive features, including temporal change rates in key sensor metrics, wear and degradation indicators and statistical to reduce noise and highlight meaningful trends.
- **Data Visualization (Responsible member(s): Taghreed - Rowan - Alaa)**
 - Generate visual analytics such as time-series graphs, correlation heatmaps, and anomaly plots to illustrate degradation behavior, sensor irregularities, and critical predictive features.

Deliverables:

- Advanced Analysis Report
- Visualizations
- Feature Engineering Summary

Week 3: *Model Development and Optimization*

- **Model Selection (Responsible member(s): Alaa -Taghreed - Nada - Rowan)**
 - Choose appropriate machine learning or deep learning models tailored to time-series forecasting, failure prediction and anomaly detection.
- **Model Training (Responsible member(s): Alaa - Taghreed -Nada - Rowan)**
 - Partition the dataset into training, validation, and testing subsets to ensure robust performance assessment.
 - Implement time-series cross-validation strategies to maintain temporal integrity and prevent information leakage.
- **Model Evaluation (Responsible member(s): Alaa - Taghreed - Nada - Rowan)**
 - Assess model effectiveness using evaluation metrics such as precision, recall, F1-score and ROC-AUC.

- **Model Optimization (Responsible member(s): Alaa - Taghreed - Nada - Rowan)**
 - Optimize hyperparameters through techniques like Grid Search or Random Search to enhance predictive performance.
 - Investigate ensemble learning approaches by combining multiple models to improve robustness and accuracy.

Deliverables:

- Model Evaluation Report
- Model Code
- Final Model

Week 4: *MLOps, Deployment, and Continuous Monitoring*

- **MLOps Setup (Responsible member(s): Taghreed - Nada)**
 - Utilize platforms such as MLflow or Kubeflow to track experiments, manage model versions, and automate training and deployment workflows.
- **Model Deployment (Responsible member(s): Taghreed - Nada)**
 - Publish the trained model as a web service or API using frameworks like Flask or FastAPI to deliver real-time failure predictions.
 - Connect the deployed model with IoT-enabled industrial systems to enable live inference and automated alerting.
- **Continuous Monitoring (Responsible member(s): Nada)**
 - Establish monitoring mechanisms to track prediction accuracy, detect model drift, and evaluate system reliability over time.
 - Develop automated retraining pipelines that incorporate newly collected data to maintain and improve model performance continuously.

Deliverables:

- Deployed Model
- MLOps Report
- Monitoring Setup